

GAINESVILLE 2 NE, AL, USA

WMO: 750030

Lat: 32.8369N

Lon: 88.1374W

Elev: 107

StdP: 14.64

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 63892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	26.5	N/A	N/A	N/A	N/A	N/A	N/A	14.5	38.2	12.9	39.2	5.6	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	95.3	N/A	93.2	N/A	91.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.6	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	12.1	10.5	9.0	DB	17.3	97.6	4.2	2.4	14.3	99.3	11.8	100.7	9.4	102.0	6.3	103.7
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	44.7	49.3	56.8	63.7	71.5	79.2	80.4	80.6	76.2	65.0	53.4	49.1
	DBStd	14.75	10.44	9.66	9.17	6.85	5.75	3.57	3.10	3.61	5.18	8.06	8.30	9.65
	HDD50	599	231	121	43	3	0	0	0	0	0	2	58	141
	HDD65	2441	632	447	278	104	19	0	0	0	1	103	357	500
	CDD50	5790	66	100	254	413	666	875	944	948	785	466	160	113
	CDD65	2158	2	6	24	64	220	425	479	483	336	102	10	7
	CDH74	19391	4	29	180	572	1841	4102	4427	4445	2836	882	56	17
	CDH80	8088	0	1	21	87	614	1904	1952	2036	1216	251	6	0
Wind	WSAvg	4.0	4.9	4.7	5.0	4.5	3.3	3.5	3.2	3.1	3.2	3.7	3.7	4.5
Precipitation	PrecAvg	51.9	5.6	5.4	5.7	4.7	3.5	3.7	4.6	3.6	3.7	3.0	4.0	5.1
	PrecMax	72.5	13.5	13.9	13.3	18.4	9.8	8.7	11.7	10.1	9.0	8.2	14.5	19.3
	PrecMin	35.7	1.1	1.5	1.5	0.1	0.1	0.6	1.1	0.4	0.7	0.0	0.7	1.9
	PrecStd	8.5	2.4	2.8	2.8	3.4	2.1	1.9	2.7	2.2	2.2	1.9	2.5	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.1	77.7	82.7	84.9	90.8	97.0	96.1	99.3	95.5	89.1	79.6	76.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	69.8	73.2	79.0	82.6	88.3	94.5	93.4	95.5	92.7	86.0	75.0	72.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	65.7	69.5	75.3	80.3	86.2	92.1	91.6	92.8	90.3	82.7	71.8	68.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	61.5	65.7	72.0	77.5	83.9	89.6	89.6	90.1	87.6	79.1	68.3	65.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	19.8	20.6	21.0	22.1	20.7	19.8	17.8	18.3	20.1	22.3	22.9	18.5
		MCDBR	22.5	23.9	24.3	24.4	23.4	23.1	20.2	22.3	23.1	25.2	25.1	20.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.326	0.333	0.364	0.400	0.437	0.482	0.526	0.524	0.439	0.373	0.346	0.324	
	taud	2.515	2.496	2.423	2.314	2.256	2.161	2.062	2.063	2.286	2.454	2.473	2.537	
	Ebn at Noon	283	293	290	282	272	258	246	245	262	274	273	278	
	Edh at Noon	27	30	35	41	44	48	53	52	40	31	28	25	
All-Sky Solar Radiation	RadAvg	855	1025	1387	1749	1930	1999	1916	1769	1574	1308	969	734	
	RadStd	70	140	126	135	134	152	114	116	125	164	98	73	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+257	+173

Nomenclature: See separate page